

Midterm Exam 3 Solutions & Rubric

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Suggested Point Distribution:

Undergraduate Total: 80 pts — Graduate Total: 100 pts

- **Exercise 1:** 20 points (5 pts for PFD, 5 pts per series region)
- **Exercise 2:** 20 points (5 pts for poles, 10 pts for residues, 5 pts for final integral)
- **Exercise 3:** 20 points (5 pts for UHP poles, 10 pts for residues, 5 pts for evaluation)
- **Exercise 4:** 20 points (10 pts for z -substitution, 10 pts for residue evaluation)
- **Exercise 5 (Grad):** 20 points (5 pts for poles/residues, 15 pts for contour setup/limit)

Exercise 1

Consider the rational function: $f(z) = \frac{2}{(z-2)(z-4)}$

- (a) Find the partial fraction decomposition of $f(z)$.
- (b) Compute the Laurent series expansion for $f(z)$ centered at $z_0 = 0$ in the three regions: (i) $|z| < 2$, (ii) $2 < |z| < 4$, (iii) $|z| > 4$.

Solution: (a) Partial Fraction Decomposition:

$$\frac{2}{(z-2)(z-4)} = \frac{A}{z-2} + \frac{B}{z-4} \implies A(z-4) + B(z-2) = 2$$

Let $z = 2 \implies A(-2) = 2 \implies A = -1$.

Let $z = 4 \implies B(2) = 2 \implies B = 1$.

$$f(z) = -\frac{1}{z-2} + \frac{1}{z-4}$$

(b) Series Expansions:

i. **Region $|z| < 2$:** Both $|z/2| < 1$ and $|z/4| < 1$.

$$-\frac{1}{z-2} = \frac{1}{2(1-z/2)} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$$

$$\frac{1}{z-4} = -\frac{1}{4(1-z/4)} = -\frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{z}{4}\right)^n = -\sum_{n=0}^{\infty} \frac{z^n}{4^{n+1}}$$

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} - \frac{1}{4^{n+1}} \right) z^n$$

ii. **Region $2 < |z| < 4$:** Here $|2/z| < 1$ and $|z/4| < 1$.

$$-\frac{1}{z-2} = -\frac{1}{z(1-2/z)} = -\frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{2}{z}\right)^n = -\sum_{n=0}^{\infty} \frac{2^n}{z^{n+1}}$$

The second term remains the same as in region (i).

$$f(z) = -\sum_{n=0}^{\infty} \frac{2^n}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{z^n}{4^{n+1}}$$

iii. **Region** $|z| > 4$: Both $|2/z| < 1$ and $|4/z| < 1$.

$$\frac{1}{z-4} = \frac{1}{z(1-4/z)} = \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{4}{z}\right)^n = \sum_{n=0}^{\infty} \frac{4^n}{z^{n+1}}$$

The first term remains the same as in region (ii).

$$f(z) = \sum_{n=0}^{\infty} \frac{4^n - 2^n}{z^{n+1}}$$

Exercise 2

Evaluate $\oint_{|z|=2} \frac{z^3 + 2z}{(z-i)^2(z+1)} dz$.

Solution: The integrand $f(z) = \frac{z^3+2z}{(z-i)^2(z+1)}$ has two singularities inside the circle $|z| = 2$: a simple pole at $z = -1$ and a pole of order 2 at $z = i$.

Residue at $z = -1$:

$$\text{Res}(f, -1) = \left. \frac{z^3 + 2z}{(z-i)^2} \right|_{z=-1} = \frac{(-1)^3 + 2(-1)}{(-1-i)^2} = \frac{-3}{1+2i+i^2} = \frac{-3}{2i} = \frac{3i}{2}$$

Residue at $z = i$: Use the derivative limit formula for $m = 2$:

$$\text{Res}(f, i) = \lim_{z \rightarrow i} \frac{d}{dz} \left[\frac{z^3 + 2z}{z+1} \right]$$

By the quotient rule:

$$\frac{d}{dz} \left[\frac{z^3 + 2z}{z+1} \right] = \frac{(3z^2 + 2)(z+1) - (z^3 + 2z)(1)}{(z+1)^2} = \frac{2z^3 + 3z^2 + 2}{(z+1)^2}$$

Evaluate at $z = i$:

$$\text{Res}(f, i) = \frac{2(i)^3 + 3(i)^2 + 2}{(i+1)^2} = \frac{-2i - 3 + 2}{2i} = \frac{-1 - 2i}{2i} = -\frac{1}{2i} - 1 = \frac{i}{2} - 1$$

Total Integral: By the Residue Theorem:

$$\oint_{|z|=2} f(z) dz = 2\pi i \left[\frac{3i}{2} + \left(\frac{i}{2} - 1 \right) \right] = 2\pi i(2i - 1) = -4\pi - 2\pi i$$

Exercise 3

Evaluate the real integral: $\int_{-\infty}^{\infty} \frac{1}{x^4 + 16} dx$.

Solution: Let $f(z) = \frac{1}{z^4 + 16}$. We integrate over the standard semicircular contour C in the upper half-plane (UHP), bounded by $[-R, R]$ and the arc C_R .

Poles in UHP: Solve $z^4 = -16 = 16e^{i\pi}$. The roots are $z_k = 2e^{i(\pi/4 + k\pi/2)}$. The two roots in the UHP are:

$$z_0 = 2e^{i\pi/4} = \sqrt{2} + i\sqrt{2} \quad \text{and} \quad z_1 = 2e^{i3\pi/4} = -\sqrt{2} + i\sqrt{2}$$

Residues: Using the $P(z)/Q'(z)$ rule, $Q'(z) = 4z^3$. Note that $z_k^4 = -16$, so $z_k^3 = -16/z_k$.

$$\text{Res}(f, z_k) = \frac{1}{4z_k^3} = \frac{z_k}{4z_k^4} = -\frac{z_k}{64}$$

$$\text{Res}(f, z_0) = -\frac{\sqrt{2} + i\sqrt{2}}{64} \quad \text{and} \quad \text{Res}(f, z_1) = -\frac{-\sqrt{2} + i\sqrt{2}}{64}$$

Sum of residues = $-\frac{2i\sqrt{2}}{64} = -i\frac{\sqrt{2}}{32}$.

Integral Evaluation: By the Residue Theorem, the integral over the closed contour is:

$$\oint_C f(z) dz = 2\pi i \left(-i\frac{\sqrt{2}}{32} \right) = \frac{2\pi\sqrt{2}}{32} = \frac{\pi\sqrt{2}}{16}$$

As $R \rightarrow \infty$, the integral over C_R vanishes because $|f(z)| \approx 1/R^4$. Thus, the real integral evaluates exactly to $\frac{\pi\sqrt{2}}{16}$.

Exercise 4

Evaluate the definite integral: $\int_0^{2\pi} \sin^4(\theta) d\theta$.

Solution: Use the substitution $z = e^{i\theta}$, meaning $d\theta = \frac{dz}{iz}$ and $\sin \theta = \frac{z - 1/z}{2i}$. The integral transforms into a contour integral over the unit circle $C : |z| = 1$:

$$\int_0^{2\pi} \sin^4(\theta) d\theta = \oint_{|z|=1} \left(\frac{z - 1/z}{2i} \right)^4 \frac{dz}{iz}$$

Expand the binomial $(z - 1/z)^4$:

$$(z - 1/z)^4 = z^4 - 4z^2(1/z) + 6z^2(1/z^2) - 4z(1/z^3) + 1/z^4 = z^4 - 4z^2 + 6 - 4/z^2 + 1/z^4$$

Notice that $(2i)^4 = 16$. Substituting this back into the integral:

$$\oint_{|z|=1} \frac{1}{16} (z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4}) \frac{dz}{iz} = \frac{1}{16i} \oint_{|z|=1} \left(z^3 - 4z + \frac{6}{z} - \frac{4}{z^3} + \frac{1}{z^5} \right) dz$$

The only term that produces a residue inside the unit circle (at $z = 0$) is the $1/z$ term. Its coefficient is 6.

$$\text{Res}(f, 0) = 6$$

Applying the Residue Theorem:

$$\int_0^{2\pi} \sin^4(\theta) d\theta = \frac{1}{16i} (2\pi i \cdot 6) = \frac{12\pi}{16} = \frac{3\pi}{4}$$

Exercise 5 (Graduate Only)

Evaluate the infinite series: $\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 4}$.

Solution: Let $f(z) = \frac{\pi \cot(\pi z)}{z^2 + 4}$. We integrate over the square contour C_N . The poles of $f(z)$ occur at $z = n$ (from the cotangent) and at $z = \pm 2i$ (from the denominator).

Residues at integers $z = n$:

$$\text{Res}(f, n) = \frac{\pi \cos(\pi z)}{\pi \cos(\pi z)(z^2 + 4)} \Big|_{z=n} = \frac{1}{n^2 + 4}$$

Residues at $z = \pm 2i$:

$$\text{Res}(f, 2i) = \frac{\pi \cot(\pi z)}{z + 2i} \Big|_{z=2i} = \frac{\pi \cot(2\pi i)}{4i} = \frac{\pi(-i \coth(2\pi))}{4i} = -\frac{\pi}{4} \coth(2\pi)$$

By symmetry (or direct calculation), $\text{Res}(f, -2i)$ yields the same result: $-\frac{\pi}{4} \coth(2\pi)$.

Applying the Residue Theorem: The integral over C_N equals $2\pi i$ times the sum of the enclosed residues:

$$\oint_{C_N} f(z) dz = 2\pi i \left[\sum_{n=-N}^N \frac{1}{n^2 + 4} - \frac{\pi}{4} \coth(2\pi) - \frac{\pi}{4} \coth(2\pi) \right]$$

Taking the limit as $N \rightarrow \infty$, we are given that the contour integral vanishes to 0:

$$0 = 2\pi i \left[\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 4} - \frac{\pi}{2} \coth(2\pi) \right]$$

Solving for the infinite sum:

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 4} = \frac{\pi}{2} \coth(2\pi)$$